

# CS & IT ENGINEERING

DISCRETE MATHS  
COMBINATORICS



Lecture No.

04



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TOPICS TO BE  
COVERED

Inclusion-exclusion

# COMBINATORICS



## Inclusion-Exclusion:

How many integers are  $\div$  by 2 or 5 in a set  $\{1 \dots 10\}$

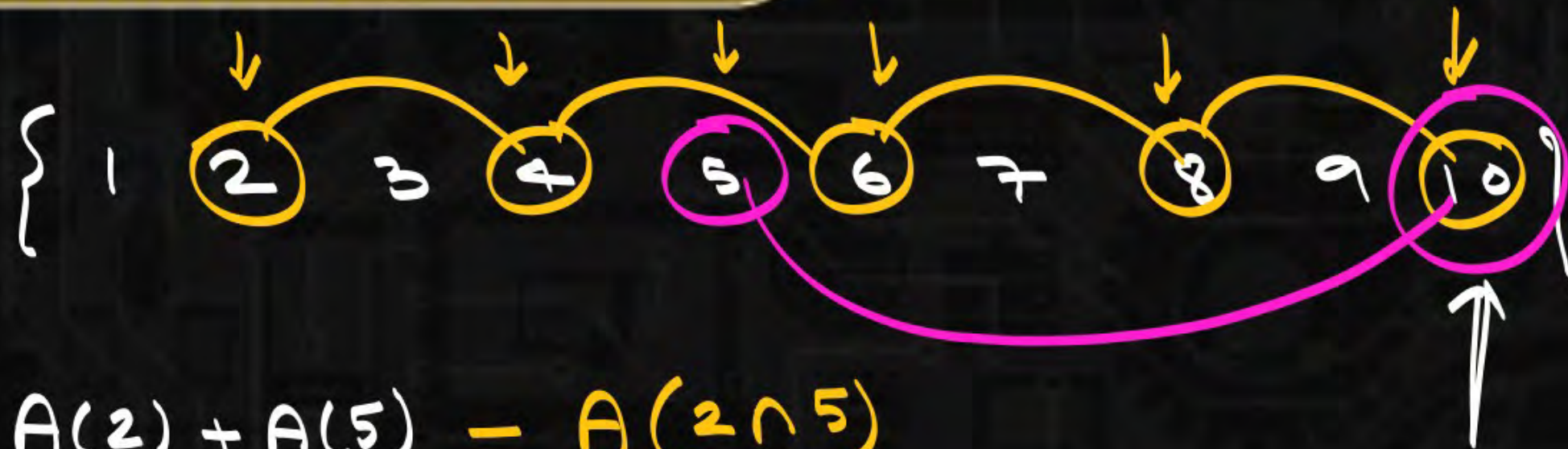
$$A(2): \text{no. of integers } \div \text{ by } 2 = \left\lfloor \frac{10}{2} \right\rfloor = 5$$

$$A(5): \text{no. of integers } \div \text{ by } 5 = \left\lfloor \frac{10}{5} \right\rfloor = 2.$$

$$\underline{\text{Ans:}} \quad 5 + 2 = 7$$



# COMBINATORICS



$$\text{Ans: } A(2) + A(5) - A(2 \cap 5)$$

$$= 5 + 2 - 1$$

$$= 7 - 1 = \underline{\underline{6}}$$

$$A(2 \cap 5) = \text{no. of elements} \\ \text{are } \div \text{ by } 2 \text{ and } 5 \\ = \left( \frac{10}{10} \right) = 1.$$

# COMBINATORICS



$$A \cup B = \underline{A} + \underline{B} - \underline{A \cap B}$$



# COMBINATORICS



$$A_1 \cup A_2 \cup A_3 = A_1 + A_2 + A_3$$

$$- A_1 \cap A_2 - A_2 \cap A_3 - A_1 \cap A_3 \\ + A_1 \cap A_2 \cap A_3$$

$$(-1)^{2+1} =$$

# COMBINATORICS



$$A_1 \cup A_2 \cup A_3 = \underline{A_1 + A_2 + A_3} - A_1 \cap A_2 - A_2 \cap A_3 - A_3 \cap A_1 + A_1 \cap A_2 \cap A_3.$$

$$(-1)^{3+1} = +ve = \sum_{i=1}^3 A_i - \sum A_i \cap A_j + \underline{A_1 \cap A_2 \cap A_3}.$$

$$A_1 \cup A_2 \cup A_3 \cup A_4 = \sum A_i - \sum A_i \cap A_j + \sum A_i \cap A_j \cap A_k - \underline{A_1 \cap A_2 \cap A_3 \cap A_4}.$$

$$(-1)^{4+1} = -ve.$$



# COMBINATORICS



$$A_1 \cup A_2 \cup A_3 \dots A_n = \sum A_i - \sum A_i \cap A_j + \sum A_i \cap A_j \cap A_k \\ \dots \dots (-1)^{n+1} A_1 \cap A_2 \cap A_3 \dots A_n$$



# COMBINATORICS



no. of integers between 1 & 500 (inclusive) that are  
 $\div 3$  or  $5$  or  $7$ .

$$\begin{aligned} A(3) \cup A(5) \cup A(7) &= A(3) + A(5) + A(7) - A(3 \cap 5) - A(5 \cap 7) - A(3 \cap 7) \\ &\quad + A(3 \cap 5 \cap 7) \\ &= \left\lfloor \frac{500}{3} \right\rfloor + \left\lfloor \frac{500}{5} \right\rfloor + \left\lfloor \frac{500}{7} \right\rfloor - \left\lfloor \frac{500}{3 \cdot 5} \right\rfloor - \left\lfloor \frac{500}{5 \cdot 7} \right\rfloor - \left\lfloor \frac{500}{3 \cdot 7} \right\rfloor \\ &\quad + \left\lfloor \frac{500}{3 \cdot 5 \cdot 7} \right\rfloor \end{aligned}$$

Ans: 221



# COMBINATORICS



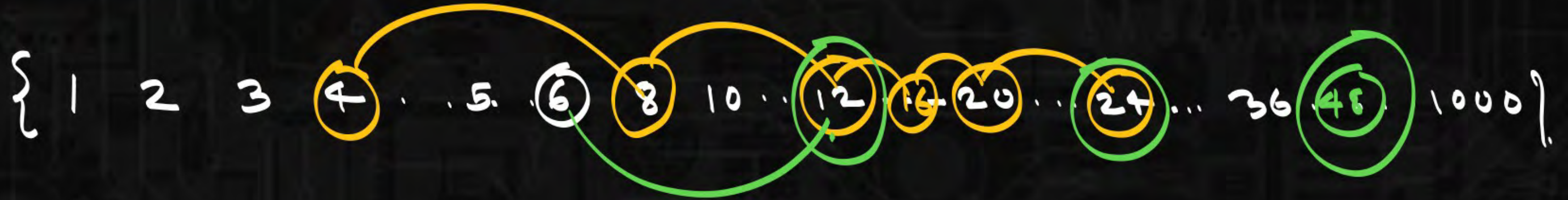
no. of integers are  $\div$  by 4 OR 6 OR 8 in a set  $\{1 \dots 1000\}$   
(inclusive)

$$A(4) \cup A(6) \cup A(8) = A(4) + A(6) + A(8) - A(4 \cap 6) - A(4 \cap 8) - A(8 \cap 6) + A(4 \cap 6 \cap 8)$$

$$= \left\lfloor \frac{1000}{4} \right\rfloor + \left\lfloor \frac{1000}{6} \right\rfloor + \left\lfloor \frac{1000}{8} \right\rfloor - \left\lfloor \frac{1000}{12} \right\rfloor - \left\lfloor \frac{1000}{8} \right\rfloor - \left\lfloor \frac{1000}{6} \right\rfloor + \left\lfloor \frac{1000}{24} \right\rfloor$$



# COMBINATORICS



$$A(4) =$$

$$A(6) =$$

